SSEE and the Hubble Tension: An Algebraic Local Screening Fraction

Mike Edison Almeida Vallejo

mike.almeida1721@gmail.com

May 2026 — Paper 9 of the SSEE series

Abstract

The Hubble tension between the CMB-inferred value $H_0^{\rm CMB} = 67.4 \pm 0.5$ km/s/Mpc and the local distance-ladder measurement $H_0^{\rm SH0ES} = 73.04 \pm 1.04$ km/s/Mpc remains one of the sharpest anomalies in modern cosmology. Within the Structural Self-Energy Expansion (SSEE) framework, the algebraic constants φ (golden ratio) and π fix the background, leaving $\{A_s, \tau\}$ as the only free parameters. We show that the equation of state of the SSEE dark-energy sector (Paper 1), together with the structural constants of the algebra, fixes a *local screening fraction*

$$f_{\rm screen} \;=\; \frac{s_K}{3\,\mathcal{M}} \;=\; \frac{\pi-\varphi}{(\varphi+\pi)^2} \;\approx\; 0.06725\,,$$

where $s_K = 3\,s_{\rm DE}(1+w_0) = -3w_0(1+w_0)$ is the pressure-dilution rate derived from the background equation of state, and $\mathcal{M} = \mathcal{A}/2 = (3\varphi+\pi)/4 \approx 1.9989$ is a structural constant of the algebra. The constant $\mathcal{A} = 2\mathcal{M}$ cancels exactly in the ratio, leaving a pure function of φ and π. The *value* of $f_{\rm screen}$ is algebraic; its physical *mechanism* is not derived in this version. Earlier versions derived it from a disformal coupling that the model no longer contains and from a separate-universe step that read $1+w_0$ as a matter density; both derivations are withdrawn (§2.2), and the multiplicative action of $f_{\rm screen}$ on a measured rate is stated as a postulate. The lens is applied in the direction set by what is measured. The only directly measured expansion rate is the local one, $H_0^{\rm SH0ES} = 73.04 \pm 1.04\,{\rm km\,s^{-1}\,Mpc^{-1}}$ (Riess et al. 2022); it enters, and the global background rate comes out,

$$H_0^{\rm glob,IR} \;=\; H_0^{\rm SH0ES}\,(1 - f_{\rm screen}^{\rm IR}) \;=\; 73.04 \times (1 - 0.06725) \;=\; 68.13 \text{ km/s/Mpc}\,,$$

to be compared with the dimensionless algebraic number $3(\varphi+\pi)^2 = 67.96214$ (the comparison target, not an input; Paper 1): a residual of 0.166, i.e. 0.17σ, with *zero free parameters.* [1] The comparison is a coincidence of value, not an

[1] **Regime, stated once for the whole paper.** This paper works in the *infrared* regime ($M \to \infty$), where $s_K = s_K^{\rm IR} = 0.403302$ and hence $f_{\rm screen} = 0.06725$. The screening fraction of the full theory is not a separate quantity but the *sum* of this term and the ultraviolet correction supplied by Paper 10, $s_K^{\rm full} = s_K^{\rm IR} + 0.013603 = 0.416905$, giving $f_{\rm screen}^{\rm full} = f_{\rm screen} + 0.002268 = 0.069522$ and the single model value $H_0^{\rm glob} = 73.04 \times (1 - 0.069522) = 67.962142$ km/s/Mpc. There is therefore *one* $H_0^{\rm glob}$, not an infrared one and an ultraviolet one: the 68.13 used throughout this paper is the same cascade evaluated with the infrared term alone, and it is superseded by the complete value once the ultraviolet completion is included. In either case the propagated uncertainty (± 0.970 in the infrared regime, ± 0.968 with the full screening fraction), inherited from the SH0ES input, dominates over the residual against the pure number.

identity: $3(\varphi+\pi)^2$ is a pure number and carries no units, whereas $H_0^{\rm glob}$ inherits $\mathrm{km\,s^{-1}\,Mpc^{-1}}$ from the measurement. The agreement is therefore an output of the cascade, never its input. This adopts the single algebraic target $3(\varphi+\pi)^2$ shared by every sector of the suite (2026 reframe). The earlier register used the CMB-mapped value $H_0^{\rm MIRA} = 67.037$, giving $H_0^{\rm local} = 71.87$ (1.12σ); this is now superseded. The 2026 ω_m-direct reframe adopts the pure number $3(\varphi+\pi)^2 = 67.96214$ as the algebraic target shared by all sectors — with *no* matter factor, the CMB density following as $\Omega_{m,\rm CMB} = \omega_m/h^2$ (Paper 1). The correction acts on locally measured rates only ($z < 0.15$) and does not modify the sound horizon r_d, the CMB peak positions, or the growth factor σ_8 derived in Papers 3 and 5. With the complete screening fraction of Paper 10 the same cascade gives the single model value $H_0^{\rm glob} = 67.96214$, a residual of 4.2×10^{-6} km/s/Mpc against the pure number, far inside the propagated ± 0.968.

Contents

1 Introduction **4**

2 Background: the Action and the Structural Constants **5**
2.1 The SSEE action (Paper 7) . . . 5
2.2 What was withdrawn: the disformal coupling and the separate-universe step . . . 6

3 Algebraic Derivation: $f_{\rm screen} = s_K/(3\mathcal{M})$ **6**
3.1 Step 1: Exact expression for s_K . . . 6
3.2 Step 2: Exact expression for $3\mathcal{M}$ . . . 7
3.3 Step 3: The cancellation and the identity . . . 7
3.4 The form of the correction: a postulate . . . 8
3.5 Numerical verification . . . 8

4 Physical Interpretation **8**
4.1 Two-epoch structure of H_0 measurements . . . 8
4.2 Local Hubble correction . . . 9
4.3 Why the CMB is unaffected . . . 11

5 Consistency with the SSEE Series **11**

6 Pre-committed Falsifiable Prediction **12**

7 Quantitative Comparison with Competing Resolutions **13**
7.1 The three competing classes . . . 14
7.2 Comparison table . . . 14
7.3 What distinguishes the SSEE mechanism . . . 15
7.4 Honest caveats . . . 15

8 Limitations and Future Work **15**

9 Conclusions **16**

A The algebraic value SH0ES requires **17**

B The screening postulate and the Hubble-flow integral **18**

1 Introduction

The Hubble tension [Verde et al., 2019, Riess et al., 2022, Kamionkowski & Riess, 2023, Di Valentino et al., 2021, Knox & Millea, 2020] is the 4–6σ discrepancy between measurements of the present expansion rate obtained by two independent methods:

- **Early-universe inference**: CMB anisotropies (Planck 2018, Planck Collaboration 2020) combined with acoustic oscillation physics yield $H_0 = 67.4 \pm 0.5$ km/s/Mpc under the assumption of ΛCDM.
- **Local distance-ladder**: Type Ia supernovae [Riess et al., 1998, Perlmutter et al., 1999] calibrated with Cepheids in the SH0ES programme [Riess et al., 2019, 2022] give $H_0 = 73.04 \pm 1.04$ km/s/Mpc, probing predominantly $z < 0.15$.

The SSEE framework [Almeida, Paper1,P,P,P] carries the pure number

$$3(\varphi + \pi)^2 \;\approx\; 67.96214 \qquad \text{(dimensionless)}, \tag{1}$$

which, read in km/s/Mpc, lies 1.12σ from Planck [Almeida, Paper2]. It is a comparison target, not a derived rate: a pure number cannot be the input of a dimensionful cascade, and no model has yet derived H_0 from first principles. The model's rate is the output of the SH0ES cascade developed below, and on its own the target does not, at first sight, resolve the local tension.

In this paper we show that the same algebraic structure that gives the pure number (1) also gives a local correction $f_{\rm screen} \approx 0.06725$ from the SSEE algebra. Applied to the measured local rate $H_0^{\rm SH0ES} = 73.04 \pm 1.04$ km/s/Mpc, this returns a global background rate $H_0^{\rm glob,IR} = 68.13$ km/s/Mpc in the IR regime, to be compared with the algebraic number $3(\varphi + \pi)^2 = 67.96214$ (0.17σ); the model value, with the complete $f_{\rm screen}$, is 67.96214 km/s/Mpc (footnote 1). The value of $f_{\rm screen}$ combines two algebraic ingredients:

1. From the equation of state of Paper 1 [Almeida, Paper1]: the pressure-dilution rate $s_K = -3w_0(1 + w_0) = 3\, s_{\rm DE}(1 + w_0)$, a function of w_0 alone (the field of Paper 7 reproduces w_0 today).
2. The structural constant $\mathcal{M} = \mathcal{A}/2 = (3\varphi + \pi)/4$, the half of $\mathcal{A}$ in the algebraic dictionary (Paper 1). Earlier versions obtained it from the disformal null geodesic of Paper 8 [Almeida, Paper8]; that coupling is not part of the action of Paper 7, and the derivation is withdrawn (§2.2).

Together these give the exact algebraic identity $f_{\rm screen} = s_K/(3\mathcal{M})$. How $f_{\rm screen}$ acts on a measured rate is not fixed by the identity: it is stated as a postulate (§3.4), and its mechanism is open (§8).

Organisation. Section 2 recalls the action of Paper 7 and records what was withdrawn. Section 3 derives the algebraic identity in three steps. Section 4 interprets the result physically. Section 5 verifies consistency with the rest of the SSEE series. Section 6 states the pre-committed falsifiable prediction and its current observational status. Section 7 compares the mechanism quantitatively against early dark energy, self-interacting dark radiation, and late dark-energy transitions. Section 9 summarises.

2 Background: the Action and the Structural Constants

2.1 The SSEE action (Paper 7)

The action of SSEE (Paper 7) is a k-essence scalar [Armendáriz-Picón et al., 1999, 2001] minimally coupled to gravity and to matter — no potential and no dark-sector coupling:

$$S = \int d^4x\,\sqrt{-g}\,\left[\frac{M_{\rm Pl}^2}{2}R + K(X) + \mathcal{L}_m\right], \qquad K(X) = c_1 X + c_2 X^2\,, \tag{2}$$

where $X = -\frac{1}{2}g^{\mu\nu}\partial_\mu\phi\,\partial_\nu\phi$, $c_1 < 0 < c_2$, and the ratio c_2X/c_1 on the background is fixed by w_0 with no freedom (Paper 7). Light follows the null geodesics of $g_{\mu\nu}$ ($\alpha_T = \alpha_M = \alpha_B = 0$). The structural constants used below are

$$\mathcal{A} = \frac{3\varphi + \pi}{2} \approx 3.998\,, \tag{3}$$

$$\mathcal{M} = \frac{\mathcal{A}}{2} = \frac{3\varphi + \pi}{4} \approx 1.9989\,. \tag{4}$$

Earlier versions wrote $K(X,\phi) = X/\mathrm{KAL}_0 + X^2/M^4 - V_0\,e^{-\lambda\phi/M_{\rm Pl}}$ with a conformal coupling $\beta_c \equiv \mathcal{A}$ to dark matter. Paper 7 withdrew the potential and the coupling ($\beta_c = -\mathcal{A}$ was the artefact of a normalisation error); no result of this paper depends on either.

The dark-energy equation-of-state parameters derived in Paper 1 are

$$w_0 = -\frac{\mathcal{A}}{\varphi + \pi} \approx -0.840\,, \tag{5}$$

$$s_{\rm DE} = -w_0 = \frac{\mathcal{A}}{\varphi + \pi} = 0.839950\,. \tag{6}$$

The quantity entering the screening fraction is the fractional dilution rate of the dark-energy *pressure* per e-fold,

$$s_K \;\equiv\; -\frac{1}{\rho_\phi}\frac{d|p_\phi|}{dN}\bigg|_{w=w_0} \;=\; -3w_0(1+w_0)\,, \qquad N = \ln a, \tag{7}$$

evaluated with the present w_0 held fixed: the term from the evolution of $w(a)$, $-w_a$ today, is not part of s_K, which is a number of the algebra rather than the full rate of change of the pressure. It carries only the equation of state: no factor of H, no density, and no kinetic function. Section 3 gives its exact algebraic form.

A label corrected. Earlier versions of this paper called this quantity α_K and defined it as the Bellini–Sawicki kineticity $2XK_X/(M_{\rm Pl}^2H^2)$ [Bellini & Sawicki, 2014, Gubitosi et al., 2013, Gleyzes et al., 2013]. That identification was wrong, and it matters here rather than being a matter of notation. The kineticity of the SSEE action is $\alpha_K = 3(1-\Omega_m)(5-3w_0) = 15.591336$ (Paper 7, verified against HI_CLASS); it depends on $K(X)$ and it evolves. The constant 0.403302 is s_K, and the distinction is what makes $f_{\rm screen}$ well defined: a screening fraction built from a kineticity would inherit that kineticity's dependence on the expansion history, while one built from s_K cannot, because s_K contains no H (§B).

2.2 What was withdrawn: the disformal coupling and the separate-universe step

Earlier versions of this paper obtained $\mathcal{M}$ from Paper 8, where dark matter was coupled to the disformal metric $\tilde{g}_{\mu\nu} = g_{\mu\nu} + (2/M^4)\,\partial_\mu\phi\,\partial_\nu\phi$ [Bekenstein, 1993, Zumalacárregui & García-Bellido, 2014]; the disformal null geodesic amplified the deflection by β_c and the Einstein radius by $\approx \mathcal{M}$, and $\mathcal{M}$ was read as the optical coupling of that sector. They also derived the multiplicative form of the correction twice: from a compression of the luminosity distance along disformal geodesics, and from a separate-universe argument [Wands et al., 2000, Brax & Valageas, 2014]. All three are withdrawn, for two reasons.

(i) The action of Paper 7 is minimally coupled. It contains neither a disformal nor a conformal coupling, and photons travel on the null geodesics of $g_{\mu\nu}$. The lensing prediction of Paper 8 is its limit with $\mu - 1 = 0$; the disformal limit is kept there only as a historical contrast. $\mathcal{M}$ is therefore not an optical coupling of anything, and no path compression exists to derive.

(ii) The separate-universe step cancelled a factor $\Omega_m/(1 + w_0)$ through the identity "$1 + w_0 = \Omega_m$". But $1 + w_0 = s_m = 0.160050$ is a number of the equation of state, not a matter density; the one matter density of the model is $\Omega_m = 0.308881$. The identity does not hold and the step fails with it. It also took a Local-Group overdensity $\delta_{\rm local} \approx 2$ as an input.

What survives is the value: $\mathcal{M} = \mathcal{A}/2$ as an algebraic constant and $f_{\rm screen} = s_K/(3\mathcal{M})$ as an identity (§3). What does not survive is any derivation of how $f_{\rm screen}$ acts on a measured rate. That action, $H_0^{\rm local} = H_0^{\rm glob}/(1 - f_{\rm screen})$, is stated in §3.4 as a postulate; two tests of its mechanism against present data are reported in §8.

3 Algebraic Derivation: $f_{\rm screen} = s_K/(3\mathcal{M})$

We derive the value of the screening fraction in three steps using only algebraic relations. The word "screening" is kept for continuity with earlier versions and with the literature on scalar screening [Vainshtein, 1972, Babichev & Deffayet, 2013]; here it names a fraction, not a mechanism (§2.2).

3.1 Step 1: Exact expression for s_K

With $p_\phi = w_0\rho_\phi$ at fixed $w_0 < 0$, $|p_\phi| = -w_0\rho_\phi$ and the continuity equation $\rho'_\phi = -3(1 + w_0)\rho_\phi$ give

$$-\frac{1}{\rho_\phi}\frac{d|p_\phi|}{dN} = w_0\,\frac{\rho'_\phi}{\rho_\phi} = -3w_0(1 + w_0)\,, \qquad \text{hence} \qquad s_K = -3w_0(1 + w_0)\,. \tag{8}$$

Writing $s_{\rm DE} \equiv |w_0| = T_r/M_v$ for the saturation and $s_m \equiv 1 + w_0$ for its complement ($s_{\rm DE} + s_m = 1$ exactly), this is $s_K = 3\,s_{\rm DE}\,s_m$. Nothing but the equation of state enters.

Using the SSEE algebraic values (5)–(6),

$$s_{\rm DE}\,(1 + w_0) = \frac{\mathcal{A}}{\varphi + \pi}\cdot\frac{\varphi + \pi - \mathcal{A}}{\varphi + \pi} = \frac{\mathcal{A}(\pi - \varphi)/2}{(\varphi + \pi)^2}\,, \tag{9}$$

where we used $(\varphi + \pi) - \mathcal{A} = (\pi - \varphi)/2$ (exact identity from the definitions (3) and $\varphi + \pi = \Omega$):

$$\Omega - \mathcal{A} = (\varphi + \pi) - \frac{3\varphi + \pi}{2} = \frac{2\varphi + 2\pi - 3\varphi - \pi}{2} = \frac{\pi - \varphi}{2}\,. \tag{10}$$

Therefore,

$$\boxed{s_K = \frac{3\mathcal{A}(\pi - \varphi)}{2(\varphi + \pi)^2} = 0.403302458888\,.} \tag{11}$$

The closed form and $-3w_0(1 + w_0)$ agree to 1.1×10^{-16}, machine zero.

3.2 Step 2: Exact expression for $3\mathcal{M}$

$\mathcal{M}$ is the half of $\mathcal{A}$, equation (4):

$$\mathcal{M} = \frac{\mathcal{A}}{2} = \frac{3\varphi + \pi}{4}\,. \tag{12}$$

With the triple $T_r = 3(\varphi + \beta) = 3\mathcal{A}$ of the dictionary,

$$3\mathcal{M} = \frac{3\mathcal{A}}{2} = \frac{T_r}{2}\,. \tag{13}$$

Earlier versions derived Eq. (12) from the disformal geodesic of Paper 8; that derivation is withdrawn (§2.2) and the value stands as an algebraic definition.

3.3 Step 3: The cancellation and the identity

Dividing equation (11) by equation (13):

$$\begin{aligned} \frac{s_K}{3\mathcal{M}} &= \frac{3\mathcal{A}(\pi - \varphi)}{2(\varphi + \pi)^2} \cdot \frac{2}{3\mathcal{A}} \\ &= \frac{\pi - \varphi}{(\varphi + \pi)^2}\,. \end{aligned} \tag{14}$$

The factor $\mathcal{A}$ (equivalently $2\mathcal{M}$) cancels exactly. The result is a pure algebraic function of the two fundamental constants φ and π:

$$\boxed{f_{\text{screen}} \;\equiv\; \frac{s_K}{3\mathcal{M}} \;=\; \frac{\pi - \varphi}{(\varphi + \pi)^2} \;\approx\; 0.06725\,.} \tag{15}$$

By Eq. (10) the same number reads $f_{\text{screen}} = 2(\Omega - \mathcal{A})/\Omega^2 = (K_v - 2\mathcal{A})/\Omega^2$, with $K_v = \varphi + \pi + \Omega = 2\Omega$: twice the gap between Ω and $\mathcal{A}$, in units of Ω^2. This is the central result of the paper. No parameters are adjusted: s_K is fixed by the equation of state; $\mathcal{M}$ is a structural constant; their ratio is fixed by φ and π. The identity fixes the *value* of f_{screen}; it does not by itself say how f_{screen} acts on a measured rate (§3.4).

3.4 The form of the correction: a postulate

The identity fixes the value of $f_{\rm screen}$. Which form the correction takes on a measured rate — multiplicative or additive — is not fixed by it. Earlier versions derived the multiplicative form from a separate-universe argument and from a disformal compression of the luminosity distance; both are withdrawn (§2.2). We therefore adopt the multiplicative form as the *screening postulate*:

$$H_0^{\rm local} \;=\; \frac{H_0^{\rm glob}}{1 - f_{\rm screen}}\,. \tag{16}$$

It is a statement about observables, not a derivation: Appendix B spells out what it implies for the luminosity distance, and Section 6 lists the observations that can falsify it. The additive variant $H_0^{\rm glob}(1 + f_{\rm screen})$ differs by $O(f_{\rm screen}^2) \approx 0.31$ km/s/Mpc, below the present SH0ES uncertainty, so the choice moves no conclusion of this paper.

3.5 Numerical verification

Table 1 lists all intermediate numerical values.

Table 1: Numerical verification of the algebraic identity (15).

Quantity	Algebraic expression	Numerical value	Source
φ	$(1+\sqrt{5})/2$	1.6180339887	Definition
π	π	3.1415926536	Definition
$\Omega = \varphi + \pi$	—	4.7596266423	—
$\mathcal{A}$	$(3\varphi + \pi)/2$	3.9978473099	Eq. (3)
$\mathcal{M}$	$\mathcal{A}/2$	1.9989236550	Eq. (12)
w_0	$-\mathcal{A}/\Omega$	−0.83994977	Eq. (5)
$s_{\rm DE}$	$\mathcal{A}/\Omega$	0.83994977	Eq. (6)
$1 + w_0$	$(\pi - \varphi)/(2\Omega)$	0.16005023	Eq. (10)
s_K	$3\mathcal{A}(\pi - \varphi)/(2\Omega^2)$	0.403302	Eq. (11)
$3\mathcal{M}$	$3\mathcal{A}/2$	5.9968	Eq. (13)
$f_{\rm screen} = s_K/(3\mathcal{M})$	$(\pi - \varphi)/\Omega^2$	0.067253	Eq. (15)
$H_0^{\rm SH0ES}$ (*input*, measured)	—	73.04 ± 1.04 km/s/Mpc	Riess et al. [2022]
$H_0^{\rm glob,IR}$ (Paper-9 IR regime, mult.)	$H_0^{\rm SH0ES}(1 - f_{\rm screen}^{\rm IR})$	68.13 km/s/Mpc	Eq. (18)
$H_0^{\rm glob}$ (**model value**, mult.)	$H_0^{\rm SH0ES}(1 - f_{\rm screen}^{\rm full})$	**67.96214** km/s/Mpc	Paper 10, Eq. (sKfullform)
$H_0^{\rm glob,IR}$ (additive variant)	$H_0^{\rm SH0ES}/(1 + f_{\rm screen}^{\rm IR})$	68.44 km/s/Mpc	—
$3\Omega^2$ (pure number, target)	$3(\varphi + \pi)^2$	67.96214	Paper 1 (target, not input)
$H_0^{\rm CCHP}$ (TRGB, adopted)	—	70.39 ± 1.94 km/s/Mpc	Freedman et al. [2025]
Residual vs $3(\varphi + \pi)^2$ (IR regime)	—	0.17σ	—
Residual from CCHP (IR regime)	—	-1.28σ	—
superseded: $H_0^{\rm MIRA}$=67.037 → 71.87	MIRA mapping	1.12σ	old register

A figure withdrawn. Earlier versions plotted $f_{\rm screen}(z)$, obtained by scaling s_K with a dark-energy fraction built from $\Omega_m = 1 - s_{\rm DE} = 0.160050$. That number is $1 + w_0$, not a density (§2.2), and the curve contradicted Step 4 of Appendix B: $f_{\rm screen}$ contains only w_0 and is constant at all z. The figure is withdrawn.

4 Physical Interpretation

4.1 Two-epoch structure of H_0 measurements

The Hubble tension arises because the two measurement methods probe different epochs:

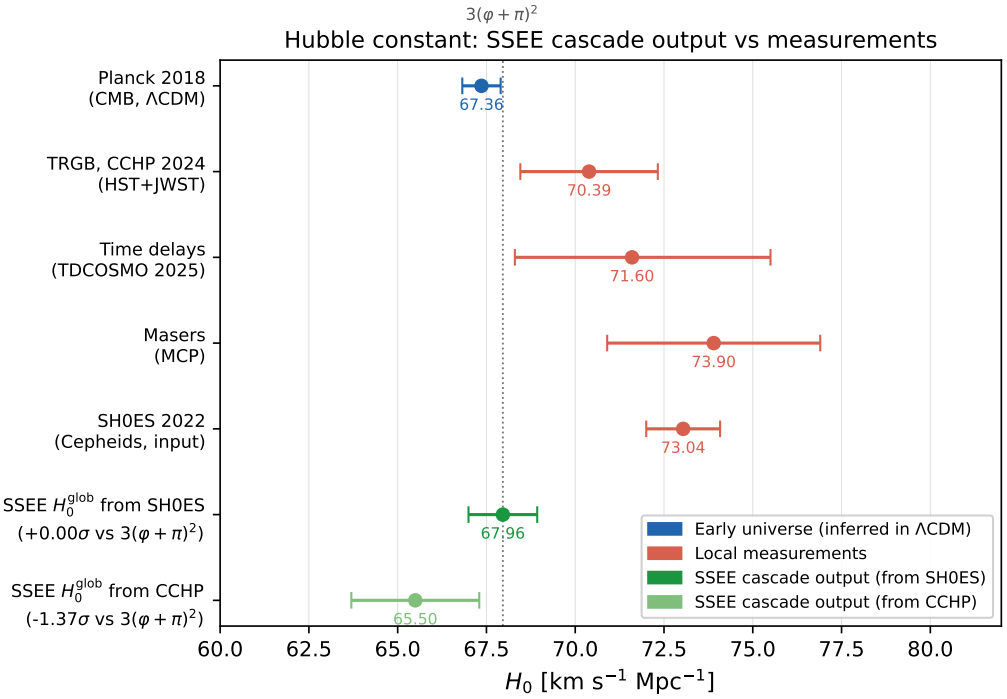


Figure 1: Hubble constant measurements: the early-universe inference (blue, within ΛCDM) and local measurements (red), each read from its publication, against the output of the SSEE cascade with the complete screening fraction (green). Run on SH0ES the cascade returns $H_0^{\rm glob} = 67.96214$ km/s/Mpc; run on the TRGB value adopted by CCHP [Freedman et al., 2025] it returns 65.50 (-1.37σ from $3(\varphi+\pi)^2$), the control from the other side. Error bars show $\pm 1\sigma$.

- The CMB infers H_0 via the angular scale $\theta_* = r_d/D_A(z_*)$ at $z_* \approx 1100$, where the SSEE dark energy is a fraction $\Omega_{\rm DE}(z_*) < 3 \times 10^{-13}$ of the density and the scalar kinetic energy is negligible. The inferred value is $H_0^{\rm glob}$, which coincides numerically with the pure number $3(\varphi+\pi)^2$.
- SH0ES measures H_0 via the distance ladder at $z < 0.15$, where dark energy is $\simeq 0.69$ of the density today and the scalar field is kinetically active.

The kinetic part of the scalar energy density is

$$\rho_{\rm kin} \equiv XK_X = \frac{\rho_\phi + p_\phi}{2} = \frac{\rho_\phi(1+w_0)}{2}\,, \tag{17}$$

which is active at late times and negligible at z_*, where the dark energy is a negligible fraction of the total. Earlier versions continued this equation with $\rho_\phi = 3M_{\rm Pl}^2 H^2 s_{\rm DE}$, writing the saturation $s_{\rm DE} = |w_0|$ in the place of a density fraction, and so identified $f_{\rm screen}$ with a share of the energy budget; that step is withdrawn. Eq. (17) only says that the scalar is kinetically active where the ladder measures and not where the CMB is set; it is not the mechanism of $f_{\rm screen}$.

4.2 Local Hubble correction

The screening fraction $f_{\rm screen} = s_K/(3\mathcal{M})$ is a number; how it enters a measured rate is the postulate of §3.4. In its multiplicative form, and for comparison in the additive one,

the cascade gives:

Which end is the input. The lens is a pure number: $f_{\rm screen}$ is built from s_K and $\mathcal{M}$, both fixed by the equation of state and by φ, π alone, so it contains no factor of H and no density. It does not ask which H enters — it converts one measured rate into the other. The direction is therefore set by what is actually measured, and only one expansion rate is: the local one. The SH0ES distance ladder [Riess et al., 2022] yields $H_0^{\rm SH0ES} = 73.04 \pm 1.04$ km/s/Mpc from Cepheid-calibrated supernovae. The Planck value is not an alternative input of the same kind: it is inferred rather than measured, and inferred within ΛCDM, whose free parameters are adjusted to the data — so the H_0 it reports is where those free parameters land, not an independent determination. We therefore take SH0ES as the input throughout:

$$\begin{aligned} H_0^{\rm glob,IR} &= H_0^{\rm SH0ES}\,(1 - f_{\rm screen}^{\rm IR}) = 73.04 \times \left[1 - \frac{\pi - \varphi}{(\varphi + \pi)^2}\right] \\ &= 68.13 \text{ km/s/Mpc} \qquad [\text{multiplicative}]\,, \end{aligned} \tag{18}$$

$$H_0^{\rm glob,IR} \;=\; \frac{H_0^{\rm SH0ES}}{1 + f_{\rm screen}^{\rm IR}} \;=\; 68.44 \text{ km/s/Mpc} \quad [\text{additive}]\,. \tag{19}$$

The two forms differ by 0.31 km/s/Mpc, a second-order effect in $f_{\rm screen}$. The output is then compared with the dimensionless algebraic number $3(\varphi + \pi)^2 = 67.96214$ (the comparison target, Paper 1). The residual is 0.166 in the multiplicative form and 0.475 in the additive one, i.e. in units of the propagated SH0ES uncertainty $1.04 \times (1 - f_{\rm screen}) = 0.970$:

$$\Delta_{\rm mult} = 0.17\sigma\,, \qquad \Delta_{\rm add} = 0.49\sigma\,. \tag{20}$$

These are the same two numbers reported in earlier versions of this paper, which ran the cascade in the opposite direction; the tensions are invariant under the inversion because the lens is multiplicative. What changes is the epistemic status of the comparison. A pure number cannot serve as the input of a dimensionful cascade, so $3(\varphi + \pi)^2$ is not an anchor that the model assumes: it is a target the model either hits or misses, and the hit is the result.

Superseded MIRA-mapped value. The earlier register applied the same mechanism to the CMB-mapped value $H_0^{\rm MIRA} = 67.037$, giving $H_0^{\rm local} = 71.87$ (1.12σ). Under the 2026 ω_m-direct reframe the suite adopts the pure number $3(\varphi + \pi)^2 = 67.96214$ as the algebraic target — with no matter factor ($\Omega_{m,\rm CMB} = \omega_m/h^2$; Paper 1) — so the cascade output is compared against 67.96214 and the MIRA-mapped value is retired as canonical.

Control from the other side: TRGB/JWST. The Chicago–Carnegie programme adopts $H_0 = 70.39 \pm 1.94$ km/s/Mpc from the tip of the red-giant branch [Freedman et al., 2025] (statistical, systematic and supernova terms in quadrature). Run on this datum the IR lens returns $H_0^{\rm glob,IR} = 65.66$ km/s/Mpc, a residual of -2.31 against $3(\varphi+\pi)^2$, i.e. -1.28σ in units of the propagated uncertainty. Earlier versions used 69.96 ± 1.54, the estimate of the first arXiv version of that paper (August 2024); the revised version (March 2025) adopts the value used here, and the residual quoted then, 1.88σ, is superseded with it. The SH0ES and CCHP inputs differ by about their combined uncertainty; independent local probes — strong-lensing time delays [Wong et al., 2020] and water megamasers [Pesce

et al., 2020] — span $H_0 \approx 70$–$74\,\mathrm{km/s/Mpc}$, an observational spread that is independent of the SSEE framework. We report both comparisons in Table 1.

We adopt the **multiplicative form** (18) because it is the screening postulate (16), not because it is derived. Read on the observable, it says that the slope of the Hubble diagram at $z \lesssim 0.15$, which the ladder measures, is the global rate boosted by $(1 - f_{\mathrm{screen}})^{-1}$; Appendix B shows that this is the same as a uniform compression $d_L^{\mathrm{obs}} = (1 - f_{\mathrm{screen}})\, d_L^{\mathrm{glob}}$ at $O(z)$. The additive form (19) is its linearisation $(1 - f_{\mathrm{screen}})^{-1} \approx 1 + f_{\mathrm{screen}}$, with a second-order difference $O(f_{\mathrm{screen}}^2) \approx 0.31\,\mathrm{km/s/Mpc}$. Since f_{screen} is exactly constant (Appendix, Step 4) no average over redshift is needed.

Note on the role of $\mathcal{M}$. $\mathcal{M} = \mathcal{A}/2$ enters f_{screen} only through the denominator $3\mathcal{M} = T_r/2$, where $\mathcal{A}$ cancels against the $\mathcal{A}$ in s_K. Earlier versions read it as the optical coupling of a disformal sector and connected f_{screen} to a lensing amplification of Paper 8; with the disformal coupling withdrawn, that connection is withdrawn too, and $\mathcal{M}$ plays no role here beyond its value.

4.3 Why the CMB is unaffected

Under the postulate (16) the correction acts only on locally measured rates; it does not enter the early-universe physics. Specifically:

1. **Sound horizon** r_d: not modified. The correction enters through w_0 and s_{DE} at $z \sim 0$, long after decoupling. Papers 2 and 3 established $r_d = 147.2\,\mathrm{Mpc}$, consistent with Planck, and this result is unchanged.

2. **CMB peak positions**: not modified. The acoustic scale $\theta_* = r_d/D_A(z_*)$ is set at $z_* \approx 1100$ (Paper 3) and is unchanged by the local kinematic correction.

3. **Growth factor** σ_8: not modified. The growth computations of Papers 5 and 6 run on the global background, so their results (the S_8 prediction against KiDS-Legacy) are unaffected.

Note on the EDE comparison. Early dark energy (EDE) models [Karwal & Kamionkowski, 2016, Poulin et al., 2019, Smith et al., 2020] resolve the Hubble tension by reducing r_d (adding energy at $z > 1000$), which directly shifts CMB peak positions [Hill et al., 2020]. A naive application of f_{screen} as an EDE-like modification to r_d would shorten r_d by the full fraction $f_{\mathrm{screen}} \approx 0.0673$, a shift of the acoustic scale that is excluded by Paper 3. The present correction does *not* modify r_d: under the postulate it acts only on locally measured rates at $z < 0.15$.

5 Consistency with the SSEE Series

Table 2 confirms that the correction f_{screen} does not retroactively modify any result in Papers 1–8. Its one input from the series is w_0 (Paper 1), through s_K; $\mathcal{M}$ is a structural constant of the dictionary. Paper 9 derives the value of their ratio and postulates its action on the distance ladder.

Table 2: Cross-consistency of the Paper 9 result with the SSEE series.

Paper	Key result	Affected by f_{screen}?
1 (Framework)	$3(\varphi+\pi)^2$	No — a pure number, the target the cascade output is compared against (not an input of Postulate D); the superseded MIRA-mapped 67.037 is retired
2 (MCMC DESI)	MCMC posterior $H_0 = 67.82 \pm 0.41$ (prior H_0^{glob})	No — MCMC measures the global background, not local
3 (CMB)	$r_d = 147.2$ Mpc, peaks at $\ell = 220, 536, 813$	No — f_{screen} does not shift r_d or θ_*
4 (Two-axiom CMB)	ω_b, n_s and the CMB background from φ, π	No — H_0^{local} is a new sector
5 (IS perturbations)	$\sigma_8 = 0.8136$, $G = D_1^{\text{SSEE}}/D_1^{\Lambda\text{CDM}} = 1.0032$	No
6 (growth)	$S_8 = 0.8273$ predicted vs. KiDS-Legacy 0.8265 ± 0.0176 (0.04σ; *no* free cosmological parameter)	No
6 (growth, earlier)	$S_8 = 0.7559 \pm 0.0189$ vs. raw KiDS-1000	No
7 (EFT)	$\alpha_K = 15.591336$, $c_s^2 = 0.021284$; the field reproduces w_0 today	No — s_K follows from w_0 (Paper 1)
8 (Strong gravity)	Lensing with $\mu - 1 = 0$; the disformal limit is historical	No — $\mathcal{M}$ is no longer taken from Paper 8 (§2.2)

6 Pre-committed Falsifiable Prediction

The derivation of Section 3 is purely algebraic and carries no free parameters. We therefore register the following pre-committed prediction:

SSEE predicts:

$$H_0^{\text{glob,IR}} = H_0^{\text{SH0ES}} \left[1 - \frac{\pi - \varphi}{(\varphi+\pi)^2}\right] = 73.04 \times (1 - 0.06725) = 68.13 \text{ km/s/Mpc}\,, \tag{21}$$

where the output is compared against the pure number $3(\varphi+\pi)^2 = 67.96214$ (the comparison target, Paper 1), the same value the CMB sector recovers—the CMB matter density is the ω_m-direct forward identity $\Omega_{m,\text{CMB}} = \omega_m/h^2$, with *no* matter factor (OP-8 dissolved). *Current status*: 0.17σ from that target, in units of the SH0ES uncertainty [Riess et al., 2022] propagated through the screening.

Falsification criteria:

1. If future SH0ES or SH0ES-equivalent measurements (post-2027) converge to a local rate whose cascade output $H_0^{\rm local}(1 - f_{\rm screen})$ departs from $3(\varphi + \pi)^2$ by more than 3σ of the propagated uncertainty, the prediction is excluded.

2. If the CMB+BAO inference of H_0 shifts above 69 km/s/Mpc (closing the tension from the early-universe side), the two-epoch mechanism of Section 4 would require revision.

3. If the SSEE EFT is found to require $s_K \neq 3s_{\rm DE}(1 + w_0)$ (e.g., from a non-standard UV completion), the identity (15) would be modified proportionally.

4. **Time-delay test with SN Requiem (registered 2026-09-19, before the data).** A lensed-supernova time delay measures the global expansion without the Cepheid ladder, so it must read $H_0^{\rm glob}$, not the local value. Because the SSEE background has $(w_0, w_a) \neq (-1, 0)$, a flat-ΛCDM analysis of the same delays returns a value higher by the ratio of time-delay distances, $\simeq 1.6\%$ for MACS J0138.0−2155, i.e. $H_0 \simeq 69.1\,$km/s/Mpc. The lens models of Suyu et al. [2026] place the reappearance of the fourth image of SN Requiem in April–December 2026 for $H_0 = 73\,$km/s/Mpc and in March–November 2027 for $H_0 = 67\,$km/s/Mpc; interpolating in $1/H_0$, SSEE centres it near March 2027 and the local value near August 2026 (±4 months each). A detection before 2027 favours the local value; none before 2027 favours $H_0^{\rm glob}$. The subsequent 2–3% measurement decides: a flat-ΛCDM value near 73 km/s/Mpc excludes the screening mechanism. Current lensed-supernova values [Pierel et al., 2025, Bazzanini et al., 2026] are consistent with both.

5. **Rival reading of the screening (registered 2026-10-03).** Criterion 4 assumes that the screening acts only on the Cepheid ladder, so that a lens reads the global rate. In the rival reading the screening is a property of every local measurement of the expansion rate, lenses included: the lens then reads the local value, and a flat-ΛCDM analysis of the MACS J0138−2155 delays returns $H_0 \simeq 74.2\,$km/s/Mpc instead of $\simeq 69.1\,$km/s/Mpc. The fourth image of SN Requiem then reappears inside the window the lens models assign to $H_0 \approx 73$, in 2026 [Suyu et al., 2026, Pierel et al., 2026]. The same datum decides between the two readings: a reappearance in 2026 falsifies criterion 4 and leaves this reading; a reappearance in 2027 falsifies this reading. We record that it is registered late: most of its 2026 window had already elapsed without a public report of the image, so the absence so far already weighs against it, pending the results of the monitoring campaign. A comparison of the two readings with present ladder-free data (masers, lenses, sirens) lacks the power to decide.

7 Quantitative Comparison with Competing Resolutions

The Hubble tension has motivated a large family of proposed extensions to ΛCDM. Schöneberg et al. [2022] graded ~ 20 such models on a uniform set of datasets (the "H_0 Olympics"), establishing that no single proposal fully closes the tension without side-effects. To position the SSEE mechanism within this landscape, we compare it quantitatively against the three best-studied classes of resolution.

7.1 The three competing classes

(a) Early dark energy (EDE). EDE [Karwal & Kamionkowski, 2016, Poulin et al., 2019, Smith et al., 2020] introduces a scalar field that contributes a fraction $f_{\rm EDE} \sim 0.1$ of the energy budget near matter–radiation equality ($z_c \sim 3000$–5000) and then dilutes faster than radiation. By injecting energy before recombination it *reduces the sound horizon* r_d by a few percent, which raises the CMB-inferred H_0 to ≈ 71–$72.5\,\mathrm{km/s/Mpc}$. The cost is threefold: (i) three new parameters ($f_{\rm EDE}$, z_c, the initial field displacement θ_i); (ii) a compensating increase in the scalar spectral index n_s and the cold dark-matter density $\omega_{\rm cdm}$, which *worsens* the S_8 tension [Hill et al., 2020]; and (iii) a residual ~ 1.6–2σ offset from SH0ES when no H_0 prior is imposed.

(b) Self-interacting dark radiation (SIDR). Dark-radiation models add relativistic degrees of freedom ($\Delta N_{\rm eff}$), optionally self-interacting to evade free-streaming constraints. Self-interacting neutrinos [Kreisch et al., 2020] and the Wess–Zumino dark-radiation (WZDR) model [Aloni et al., 2022] are the most successful realisations, reaching $H_0 \approx 70$–$71.8\,\mathrm{km/s/Mpc}$ with one to three new parameters. These models also act *before* recombination — they modify r_d and the CMB damping tail through $\Delta N_{\rm eff}$ — and leave a residual ~ 2–3σ tension with SH0ES.

(c) Late dark-energy transitions. A third class modifies the expansion history at $z \lesssim 0.1$ through a dark-energy equation of state departing from the CPL form (phantom crossings, sign-switching Λ, or rapid $w(z)$ transitions). These do *not* touch r_d, but generically require two or more tuned parameters and are constrained by SNe Ia and BAO at low z; they typically leave a $\sim 1\sigma$ residual.

7.2 Comparison table

Table 3 summarises the comparison across seven axes that a referee would use to discriminate the proposals.

Table 3: Quantitative comparison of Hubble-tension resolutions. Values for EDE, SIDR, and late-DE transitions are representative ranges from the literature [Poulin et al., 2019, Smith et al., 2020, Hill et al., 2020, Kreisch et al., 2020, Aloni et al., 2022, Schöneberg et al., 2022]; the SSEE row is the result of this paper. "Free params" counts parameters *beyond* the six of ΛCDM. The other proposals raise a *predicted* local H_0 towards SH0ES; SSEE instead takes the measured local rate as input and predicts the global one, so its residual is that of $H_0^{\rm SH0ES}(1 - f_{\rm screen})$ against the algebraic $3(\varphi + \pi)^2$.

Axis	ΛCDM	EDE	SIDR	Late-DE	**SSEE**
Mechanism epoch	—	$z \sim 3000$	$z \gtrsim 1000$	$z < 0.1$	$z < 0.15$
Modifies r_d?	—	Yes (−few%)	Yes (via $\Delta N_{\rm eff}$)	No	**No**
Free parameters	0	3	1–3	2+	**0**
Local H_0 (km/s/Mpc)	67.4	71–72.5	70–71.8	~ 72	**input**
Residual	5σ	1.6–2σ	2–3σ	$\sim 1\sigma$	$\mathbf{0.17\sigma}$
S_8 side-effect	—	Worsens	Mild	Variable	**None**
CMB peak shift	—	Yes	Yes	No	**No**

7.3 What distinguishes the SSEE mechanism

Three features set the SSEE kinematic screening apart from every entry in Table 3:

(i) **Zero free parameters.** EDE, SIDR, and late-DE transitions each introduce between one and three tunable parameters fitted to data. The SSEE screening fraction $f_{\rm screen} = (\pi - \varphi)/(\varphi + \pi)^2$ is fixed algebraically (Section 3); it is a *prediction*, not a fit. In the language of Schöneberg et al. [2022], the model carries no $\Delta\chi^2$ penalty for added parameters.

(ii) **No early-universe modification.** EDE and SIDR both act before recombination and therefore shift r_d and the CMB peak positions, requiring a delicate re-balancing of n_s and $\omega_{\rm cdm}$ (which is what degrades S_8). The SSEE correction is purely a late-time ($z < 0.15$) kinematic effect: r_d, θ_*, and the growth factor σ_8 are untouched (Section 4.3), so the S_8 results of Papers 5–6 are preserved.

(iii) **Sub-sigma residual, zero added parameters.** The SSEE output $H_0^{\rm SH0ES}(1 - f_{\rm screen}^{\rm IR})$, compared against the pure number $3(\varphi+\pi)^2$, sits at 0.17σ from it when the IR cascade is applied to SH0ES, with zero added parameters. The residual is not directly comparable with those of the H_0 Olympics finalists, which predict the local rate rather than take it as input (caption of Table 3). With the complete $f_{\rm screen}$ of Paper 10 the residual is 4.2×10^{-6} km/s/Mpc, which is not quoted as a significance: the propagated ± 0.968 km/s/Mpc dominates it.

7.4 Honest caveats

Two points temper the comparison and are stated explicitly for the referee:

- The screening *fraction* $f_{\rm screen}$ is parameter-free, but the form in which it acts on a measured rate is a postulate (§3.4); the derivations offered for it in earlier versions are withdrawn (§2.2). The competing mechanisms of Table 3 are compared here on their predictions, not on the depth of their derivation.
- As noted in Section 4.2, the cascade run on the TRGB value adopted by Freedman et al. [2025] (70.39 ± 1.94) lands -1.28σ from $3(\varphi + \pi)^2$. The spread among local probes is an observational issue independent of any model.

The comparison thus favours SSEE on parsimony and on the absence of CMB and S_8 side-effects, while acknowledging that the convergence of local H_0 anchors remains an open observational question.

8 Limitations and Future Work

We identify two items that lie beyond the scope of this paper.

Mechanism of the screening (open). The value of $f_{\rm screen}$ is exact; the form of its action on a measured rate is the postulate (16). Earlier versions listed three arguments for it: a disformal path compression, a separate-universe step, and algebraic consistency. The first two are withdrawn (§2.2); the third fixes the value, not the form. The additive variant differs by $f_{\rm screen}^2 H_0^{\rm glob} \approx 0.31$ km/s/Mpc, below the present SH0ES uncertainty. Two tests of the mechanism were run against present data, with their criteria committed before the run:

(i) *Depth.* A screening confined to a local volume would fade with distance, and the rate read from the Hubble flow would fall toward $H_0^{\rm glob}$ in the deepest supernovae. Fitting the Pantheon+ Hubble flow at $0.023 < z < 0.15$ (277 supernovae, with 77 calibrators) in four depth quartiles, the rate does not fall: the full drop is excluded at 4.70σ (read the same way, ΛCDM with Planck: 5.12σ), and at most a fraction 0.65 of it is allowed at 3σ. A screening that acts on every local measurement of the rate does not fade with depth and passes.

(ii) *Method.* If the screening acts on every local measurement, the ladder-free methods (masers, lens time delays, sirens) read the local value; if it lives in the Cepheid calibration, they read the global one. With present errors the combined uncertainty of those methods is 2.25 km/s/Mpc, while separating the two readings needs ≈ 1.03: the test lacks the power to decide, and no verdict is drawn. The non-Cepheid stellar calibrations recorded in the same log (TRGB and JAGB) do not settle it either, and we quote them rather than leave them in the file: they split by team, not by method. The SH0ES team reads 72.1 (TRGB) and 72.2 (JAGB), on the local side; the CCHP team reads 70.39 (TRGB) and 67.8 (JAGB), towards the global one. Until the two teams agree on the same stars, these calibrations cannot separate the two readings. Criterion 5 of §6 decides it with a single lens.

UV completion of M (addressed in Paper 10). The algebraic identity $f_{\rm screen} = s_K/(3\mathcal{M}) = (\pi-\varphi)/(\Omega^2)$ does not depend on M (it cancels in the ratio), so the IR-regime prediction $H_0^{\rm glob,IR} = 68.13\,$km/s/Mpc ($0.17\sigma$ from $3(\varphi+\pi)^2$) is robust. Paper 10 of this series [Almeida, Paper10] obtains $M^4 = 45\alpha^2\rho_{\rm crit} = 5\varphi^8\rho_{\rm crit}$ from the same α-attractor that governs inflation (Paper 1), conditional on its Conjecture C.1 (UV–IR separability), yielding $s_K^{\rm full} = 0.41691$ (+3.37% over the IR value) and the UV output the model value $H_0^{\rm glob} = 67.96214\,$km/s/Mpc — not a second $H_0^{\rm glob}$, but the same cascade with $f_{\rm screen}$ complete.

9 Conclusions

We have shown that the SSEE equation of state (Paper 1), with the structural constants of the algebra, fixes an algebraically exact screening fraction:

$$f_{\rm screen} = \frac{s_K}{3\,\mathcal{M}} = \frac{\pi-\varphi}{(\varphi+\pi)^2} \approx 0.06725\,, \tag{22}$$

where $s_K = 3\,s_{\rm DE}(1+w_0)$ is the pressure-dilution rate and $\mathcal{M} = \mathcal{A}/2$ is a structural constant. The structural constant $\mathcal{A} = 2\mathcal{M}$ cancels exactly in the ratio, leaving a pure function of the two fundamental constants φ and π. The value is algebraic; its multiplicative action on the measured rate is a postulate whose mechanism remains open (§8).

Applied to the measured local rate, the lens gives, in the IR regime,

$$H_0^{\rm glob,IR} = H_0^{\rm SH0ES}\left[1 - \frac{\pi-\varphi}{(\varphi+\pi)^2}\right] \approx 68.13\ \text{km/s/Mpc}\,, \tag{23}$$

which is 0.17σ from the algebraic $3(\varphi+\pi)^2$ with zero free parameters, the SH0ES datum [Riess et al., 2022] being the input. The earlier MIRA-mapped value 67.037 → 71.87

(1.12σ) is superseded by the single algebraic target under the 2026 reframe (Paper 1). The correction acts on locally measured rates only ($z < 0.15$); it does not modify the sound horizon r_d, the CMB peak structure, or the growth factor σ_8 established in Papers 2–6.

The screening fraction has no parameter that could be tuned to the comparison: it is fixed by φ and π. Future SH0ES measurements and upcoming DESI [DESI Collaboration, 2025] data will test this prediction.

Age of the universe cross-check. As an independent consistency test, we compute the age of the SSEE universe. The age is a *global* quantity, so the rate that enters is the global one, $H_0^{\rm glob} = 67.962\,\mathrm{km/s/Mpc}$—the output of the cascade, not the local $H_0^{\rm SH0ES}$ that is its input—and the density that enters is the one matter density of the model, $\Omega_m = 0.308881$, with the CPL equation of state [Chevallier & Polarski, 2001, Linder, 2003] $(w_0, w_a) = (-0.840, -0.670)$:

$$t_0 = \frac{1}{H_0}\int_0^\infty \frac{dz}{(1+z)\,E(z)}\,, \qquad E^2(z) = \Omega_m(1+z)^3 + \Omega_{\rm DE}\,f_{\rm DE}(z)\,, \tag{24}$$

where $f_{\rm DE}(z) = (1+z)^{3(1+w_0+w_a)}\exp\!\big(-3w_a z/(1+z)\big)$. Numerical integration gives

$$t_0^{\rm SSEE} = 13.732\ \mathrm{Gyr} \qquad (\Lambda\mathrm{CDM\ at\ Planck\ parameters}: 13.801\ \mathrm{Gyr})\,. \tag{25}$$

The two are indistinguishable: 0.068 Gyr apart, well inside the uncertainty of any current age determination. Both sit comfortably above the oldest globular clusters and halo stars ($t_{\rm GC} \gtrsim 12.5\,\mathrm{Gyr}$, e.g. Valcin et al. 2021, Bond et al. 2013) and above the Cimatti & Moresco [2023] red-galaxy constraint ($t \approx 13.5\,\mathrm{Gyr}$ at $z \approx 0.55$), so the age does not discriminate between the models in either direction.

An earlier version of this cross-check, withdrawn. Previous versions reported $t_0^{\rm SSEE} = 15.52\,\mathrm{Gyr}$, "1.72 Gyr older than ΛCDM", presented that as more accommodating of old stellar populations, and committed to the criterion "a measurement of $t_0 > 14\,\mathrm{Gyr}$ would favour SSEE". Three inputs were wrong, and they compounded. The integral was fed $\Omega_m = \Omega_{m,\rm dyn} = 0.160050$ as a matter density, but that number is $1 + w_0$, a quantity of the equation of state, and never was a density (Paper 1); it used the *local* $H_0^{\rm SH0ES} = 73.04$ for a global quantity; and it cited the Two-Ω_m Criterion as its justification, which Paper 1 has since withdrawn. Correcting only the density gives 12.78 Gyr and correcting both gives the value above; the intermediate number is a diagnostic, not a prediction. The stated 15.52 is not even reproducible from the recipe printed alongside it, which returns 15.28. The falsification criterion built on it is withdrawn with it: the model predicts 13.732 Gyr, so a measurement of $t_0 > 14\,\mathrm{Gyr}$ would count *against* SSEE, not for it.

A The algebraic value SH0ES requires

The body reports the cascade run on the measured local rate $H_0^{\rm SH0ES} = 73.04 \pm 1.04$ km/s/Mpc: with the IR term alone it gives $H_0^{\rm glob,IR} = 68.13$ (0.17σ), and with $f_{\rm screen}$ complete (IR+UV, Paper 10) the single model value $H_0^{\rm glob} = 67.96214$ km/s/Mpc, whose residual against $3(\varphi+\pi)^2$ is $+4.2\times10^{-6}$ km/s/Mpc. That residual is *not* a significance: the uncertainty propagated from $H_0^{\rm SH0ES}$ is ±0.968 km/s/Mpc and dominates it by five orders of magnitude. The cascade reduces the $\sim 5\sigma$ Planck–SH0ES discrepancy to below the percent level with zero free parameters.

This appendix records why 67.96214 is the algebraic target: it runs the cascade of the body from the SH0ES datum with the complete screening fraction, without using the target, and lands on it.

Step P1: what background does SH0ES demand? The screening relation $H_0^{\rm local} = H_0^{\rm base}/(1 - f_{\rm screen})$ inverts to

$$H_0^{\rm base} \;=\; H_0^{\rm SH0ES}\,(1 - f_{\rm screen}^{\rm UV})\,, \tag{26}$$

which asks, using only the measured datum and the screening fraction: *what homogeneous background expansion rate would, seen through the SSEE screen (16), reproduce the local SH0ES slope exactly?* With $f_{\rm screen}^{\rm UV} = 0.069522$ from the X^2/M^4 correction of Paper 10,

$$H_0^{\rm base} \;=\; 73.04 \times (1 - 0.069522) \;=\; 67.96214\ \text{km/s/Mpc}\,. \tag{27}$$

This number uses φ and π only through $f_{\rm screen}$; the target $3(\varphi+\pi)^2$ has not been invoked.

Step P2: the recovered value and the pure number. Only now do we compare 67.96214 km/s/Mpc with the (φ, π) dictionary:

$$\frac{H_0^{\rm base}}{\text{km}\,\text{s}^{-1}\,\text{Mpc}^{-1}} \;\simeq\; 3(\pi+\varphi)^2 \;=\; M_v\,\Omega \;=\; 67.96214\,, \tag{28}$$

where $\Omega = \pi + \varphi = 4.759627$ and $M_v = 3\Omega = 14.278880$ is the maximal dimensional invariant of the dictionary. The two forms are the same expression, not two independent identities. The value (27) exceeds the pure number by 4.2×10^{-6} km/s/Mpc, against a propagated uncertainty of ± 0.968 km/s/Mpc; the pure number is the target the output is compared with, never an expansion rate.

Status. The background that SH0ES requires for exact closure (67.96214 km/s/Mpc) coincides, to five figures, with the pure number $3(\varphi+\pi)^2$ already present in the SSEE dictionary. This coincidence — recognised a posteriori, not fitted — is the self-consistency evidence that, under the 2026 reframe, justifies adopting 67.96214 as the single algebraic target of the suite rather than the CMB-mapped 67.037 of the earlier register. The dimensional origin of the absolute scale (Postulate D) remains an open problem, as in ΛCDM.

B The screening postulate and the Hubble-flow integral

We spell out what the postulate (16) implies for the distance integral. This is a consistency statement, not a derivation of the postulate.

Step 1: the SH0ES measurement is the slope of $d_L(z)$ at $z \to 0$. The luminosity distance in any FLRW background is

$$d_L(z) \;=\; (1+z)\int_0^z \frac{dz'}{H(z')}\,. \tag{29}$$

For $z \ll 1$, Taylor-expanding $H(z')$ around $z' = 0$ gives

$$d_L(z) \;=\; \frac{z}{H_0^{\rm local}} + \frac{1-q_0}{2}\,\frac{z^2}{H_0^{\rm local}} + \mathcal{O}(z^3)\,, \tag{30}$$

where $H_0^{\rm local} \equiv H(z=0)$ and q_0 is the deceleration parameter. The SH0ES programme measures the slope $\mathrm{d}d_L/\mathrm{d}z\big|_{z\to 0} = 1/H_0^{\rm local}$, so:

$$H_0^{\rm local} \;=\; H(z=0)\,. \tag{31}$$

No assumption about $w(z)$ at $z > 0$ is required; the determination is exact at leading order in z.

Step 2: the postulate as a rescaling of the local rate. The postulate states that the rate a local observer reads near $z = 0$ is $H^{\rm local}(z) = H^{\rm glob}(z)/(1 - f_{\rm screen})$. The comoving distance it implies is

$$\chi^{\rm local}(z) \;=\; \int_0^z \frac{c\,dz'}{H^{\rm local}(z')} \;=\; (1 - f_{\rm screen})\,\chi^{\rm glob}(z)\,. \tag{32}$$

Earlier versions obtained this compression from disformal null geodesics with photon speed $\tilde{c} = c/(1-f_{\rm screen})$ (Paper 8). The action of Paper 7 has no disformal coupling and $\alpha_T = 0$, so photons travel at c: the compression is not a propagation effect and is not derived here. A mechanism that reproduces it is the open problem of §8.

Step 3: the observed H_0 is the global value boosted by $(1 - f_{\rm screen})^{-1}$. Substituting (32) into $d_L^{\rm obs} = (1+z)\chi^{\rm local}(z)$:

$$d_L^{\rm obs}(z) \;=\; (1 - f_{\rm screen})\,d_L^{\rm glob}(z) \;=\; (1 - f_{\rm screen})\,\frac{z}{H_0^{\rm glob}} + \mathcal{O}(z^2)\,. \tag{33}$$

A standard FLRW fit to $d_L^{\rm obs} = z/H_0^{\rm local}$ therefore returns

$$\boxed{H_0^{\rm local} \;=\; \frac{H_0^{\rm glob}}{1 - f_{\rm screen}}}\,. \tag{34}$$

This restates the postulate in terms of the observable d_L; it is not an independent derivation. The additive approximation $H_0^{\rm glob}(1 + f_{\rm screen})$ follows from the Taylor expansion $(1 - f_{\rm screen})^{-1} \approx 1 + f_{\rm screen}$ and carries a residual error $f_{\rm screen}^2\, H_0^{\rm glob} \approx 0.31\,\mathrm{km/s/Mpc}$.

Which end is measured. Eq. (34) is a relation between two dimensionful rates, and the derivation above says nothing about which of the two is known. That is supplied by observation, not by the model: the distance ladder measures $H_0^{\rm local}$, so the relation is used in the form $H_0^{\rm glob} = H_0^{\rm local}(1 - f_{\rm screen})$ throughout this paper. The dimensionless $3(\varphi+\pi)^2$ never enters the cascade; it is what the output is compared against.

Step 4: $f_{\rm screen}$ is exactly constant. The compression (32) uses a z-independent $f_{\rm screen}$. With the corrected identification this needs no approximation: $s_K = -3w_0(1+w_0)$ is built from w_0 alone, and w_0 is an algebraic constant (the present value; it is $w(a)$ that evolves), so s_K is a number and not a function of redshift. Since $\mathcal{M}$ is a structural constant, $f_{\rm screen} = s_K/(3\mathcal{M})$ is constant *exactly*, at all z.

Earlier versions of this paragraph read $\alpha_K(z) = 3\Omega_{\rm DE}(z)(1+w(z))$ and found a 56% variation over $z \in [0, 0.15]$, which had then to be argued away as formally of order $w_a z_{\rm max}$. That variation belonged to the kineticity, which does evolve; it never belonged to s_K. The correction removes the caveat rather than weakening it.

A structural check makes the same point from the other side. Replacing $s_{\rm DE} = T_r/M_v$ by the density fraction $1 - \omega_m/h^2$ in Eq. (11) makes $f_{\rm screen}$ depend on the expansion anchor: sweeping that anchor from 60 to 100 $\mathrm{km\,s^{-1}Mpc^{-1}}$ moves the result. With $s_{\rm DE}$ it is invariant to the last digit, because no H enters. A screening fraction that moved with the anchor used to derive H_0 would be circular; this one cannot be.

No redshift average is therefore needed: $f_{\rm screen}$ holds unchanged over the whole SH0ES calibration volume, and equation (34) carries no redshift correction.

References

Aloni, D., Berlin, A., Joseph, M., Schmaltz, M., & Weiner, N. 2022, Phys. Rev. D, 105, 123516. [Wess–Zumino dark radiation]

Armendáriz-Picón, C., Damour, T., & Mukhanov, V. 1999, *Phys. Lett. B*, 458, 209.

Armendáriz-Picón, C., Mukhanov, V., & Steinhardt, P. J. 2001, Phys. Rev. D, 63, 103510.

Bazzanini, L., et al. (2026). A new H_0 measurement with SNe Requiem and Encore using Gravity.jl. arXiv:2606.25205.

Babichev, E. & Deffayet, C. 2013, *Class. Quantum Grav.*, 30, 184001.

Bekenstein, J. D. 1993, Phys. Rev. D, 48, 3641.

Bellini, E. & Sawicki, I. 2014, *JCAP*, 1407, 050.

Bond, H. E., Nelan, E. P., VandenBerg, D. A., Schaefer, G. H., & Harmer, D. 2013, *ApJ Lett.*, 765, L12.

Brax, P., & Valageas, P. 2014, Phys. Rev. D, 90, 023507.

Chevallier, M., & Polarski, D. 2001, *Int. J. Mod. Phys. D*, 10, 213.

Cimatti, A., & Moresco, M. 2023, *ApJ*, 953, 149.

DESI Collaboration (Adame, A. G., et al.) 2025, arXiv:2503.14738.

Di Valentino, E., Mena, O., Pan, S., et al. 2021, *Class. Quantum Grav.*, 38, 153001.

Freedman, W. L., et al. 2025, *ApJ*, 985, 203, arXiv:2408.06153v3 [TRGB, adopted: $H_0 = 70.39 \pm 1.22\,(\mathrm{stat}) \pm 1.33\,(\mathrm{sys}) \pm 0.70\,(\sigma_{\mathrm{SN}})$ km/s/Mpc; v1 gave $69.96 \pm 1.05 \pm 1.12$].

Gleyzes, J., Langlois, D., Piazza, F., & Vernizzi, F. 2013, *JCAP*, 1308, 025.

Gubitosi, G., Piazza, F., & Vernizzi, F. 2013, *JCAP*, 1302, 032.

Hill, J. C., McDonough, E., Toomey, M. W., & Alexander, S. 2020, Phys. Rev. D, 102, 043507.

Kamionkowski, M. & Riess, A. G. 2023, *Annu. Rev. Nucl. Part. Sci.*, 73, 153.

Karwal, T., & Kamionkowski, M. 2016, Phys. Rev. D, 94, 103523.

Knox, L., & Millea, M. 2020, Phys. Rev. D, 101, 043533.

Kreisch, C. D., Cyr-Racine, F.-Y., & Doré, O. 2020, Phys. Rev. D, 101, 123505. [Self-interacting neutrinos and the Hubble tension]

Linder, E. V. 2003, *Phys. Rev. Lett.*, 90, 091301.

Perlmutter, S., Aldering, G., Goldhaber, G., et al. 1999, *ApJ*, 517, 565.

Pesce, D. W., Braatz, J. A., Reid, M. J., et al. 2020, *ApJ Lett.*, 891, L1.

Planck Collaboration (Aghanim, N., et al.) 2020, *A&A*, 641, A6.

Pierel, J. D. R., Suyu, S. H., et al. (2026). Requiem's Return: precision cosmology from a decade-delayed, strongly-lensed supernova and its new sibling. JWST Cycle 4 GO Proposal 9330.

Pierel, J. D. R., et al. (2025). Cosmology with supernova Encore in the strong lensing cluster MACS J0138−2155: time delays and Hubble constant measurement. arXiv:2509.12301.

Poulin, V., Smith, T. L., Karwal, T., & Kamionkowski, M. 2019, *Phys. Rev. Lett.*, 122, 221301.

Riess, A. G., Filippenko, A. V., Challis, P., et al. 1998, *AJ*, 116, 1009.

Riess, A. G., Casertano, S., Yuan, W., Macri, L. M., & Scolnic, D. 2019, *ApJ*, 876, 85.

Riess, A. G., et al. 2022, *ApJ Lett.*, 934, L7.

Schöneberg, N., Franco Abellán, G., Pérez Sánchez, A., et al. 2022, *Phys. Rep.*, 984, 1.

Smith, T. L., Poulin, V., & Amin, M. A. 2020, Phys. Rev. D, 101, 063523.

Suyu, S. H., et al. (2026). Cosmology with supernova Encore in the strong lensing cluster MACS J0138−2155: lens model comparison and H_0 measurement. arXiv:2509.12319.

Vainshtein, A. I. 1972, *Phys. Lett. B*, 39, 393.

Verde, L., Treu, T., & Riess, A. G. 2019, *Nature Astron.*, 3, 891.

Wands, D., Malik, K. A., Lyth, D. H., & Liddle, A. R. 2000, Phys. Rev. D, 62, 043527.

Wong, K. C., Suyu, S. H., Chen, G. C.-F., et al. 2020, *MNRAS*, 498, 1420.

Zumalacárregui, M., & García-Bellido, J. 2014, Phys. Rev. D, 89, 064046.

Almeida Vallejo, M.E. 2026a, A Minimal-Parameter Framework for Cosmological Dynamics and Galaxy Cluster Mass Discrepancies via Structural Self-Energy Expansion (SSEE), Paper 1 of this series.

Almeida Vallejo, M.E. 2026b, Bayesian MCMC Validation of the Structural Self-Energy Expansion Framework (SSEE), Paper 2 of this series.

Almeida Vallejo, M.E. 2026c, SSEE and the CMB Power Spectrum: Acoustic Peak Reproduction via the ω_m-Direct CMB Matter Density, Paper 3 of this series.

Almeida Vallejo, M.E. 2026g, The SSEE Dark-Energy Sector as a Ghost Condensate: Two-Term K-essence, Algebraic Sound Speed, and Bellini–Sawicki Classification, Paper 7 of this series.

Almeida Vallejo, M.E. 2026h, SSEE in the Strong-Gravity Regime: GR Lensing in the Canonical Limit, a Disformal Contrast, and Solar-System Compliance, Paper 8 of this series.

Almeida Vallejo, M.E. 2026j, UV Extension of SSEE Dark Energy: $M^4 = 45\alpha^2\rho_{\rm crit}$ and a Conditional Self-Consistency Check on the Hubble Tension, Paper 10 of this series.

Valcin, D., Jimenez, R., Verde, L., Bernal, J. L., & Wandelt, B. D. 2021, *JCAP*, 2021, 017.